

Shifan Xu

PH.D. CANDIDATE IN APPLIED PHYSICS

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Education

Yale University

PHD CANDIDATE

New Haven, CT, USA

Sep. 2021 - present

- Advisor: Dr. Yongshan Ding, Dr. Steven M. Girvin
- Major: Applied Physics

University of Chicago

VISITING PH.D. STUDENT

Chicago, IL, USA

Jan. 2024 - May 2024

Nanjing University

BS UNDERGRADUATE DEGREE

Nanjing, China

Sep. 2017 - Jun. 2021

- Major: Physics

University of California, Berkeley

VISITING UNDERGRADUATE STUDENT

Berkeley, CA, USA

Jul. 2019 - Jan. 2020

Professional Experience

2025 **Graduate Student Intern**, Pacific Northwest National Laboratory (PNNL), Richland

2023-2025 **Graduate Research Assistant**, Dept. Applied Physics and Dept. Computer Science, Yale University

2022-2023 **Graduate Teaching Assistant**, Dept. Applied Physics and Dept. Computer Science, Yale University

2019-2021 **Undergraduate Research Assistant**, Dept. Engineering, Nanjing University

2018-2021 **Undergraduate Research Assistant**, Dept. Physics, Nanjing University

Research Experience

Yale University - Applied Physics Department

ADVISOR: DR. YONGSHAN DING

New Haven, CT, USA

Sep. 2021 - Present

- Architectural Improvements in Quantum Random Access Memory
- Practical Quantum Memory Architecture in NISQ and Fault-tolerant Era

Pacific Northwest National Laboratory (PNNL)

ADVISOR: DR. ANG LI

Richland, WA, USA

Jun. 2025 - Aug. 2025

- Project: Quantum Secure Query

Nanjing University - Physics Department

ADVISOR: DR. TIANJUE MIN

Nanjing, China

2021

- Graduation Thesis Project: "Analysis of Background Noise in $p\bar{p}$ Collision Experiments Based on the K-Nearest Neighbors (KNN) Algorithm"

UC San Diego - Department of NanoEngineering

ADVISOR: DR. SHENG XU

(remote) San Diego, CA, USA

2020

- Project: Circuit Design for Flexible Ultrasonic Array

Nanjing University - Engineering Department

ADVISOR: DR. XIN LI, DR. MINGHUI LU

Nanjing, China

2019-2021

- Project: Preparation and Design of Auxetic Metamaterials

Nanjing University - Physics Department

SUPERVISOR: PROF. SIHUI WANG

Nanjing, China

2018-2019

- Project: Metal Powder Spectra Analysis

Publications

PUBLISHED

Shifan Xu, Kun Liu, Patrick Rall, Zhiyang He, and Yongshan Ding. “Distilling Magic States in the Bicycle Architecture.” In Proceedings of the 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (ISCA), pp. 2081–2095. 2026. DOI: 10.1109/ISCA66397.2026.00148. arXiv:2602.20546.

Kun Liu, Takahiro Tsunoda, Sophia H. Xue, Evan McKinney, Zeyuan Zhou, **Shifan Xu**, Robert J. Schoelkopf, and Yongshan Ding. “Efficient Routing of Quantum LDPC Codes on Programmable 2D Toric Architectures.” In IEEE International Conference on Quantum Computing and Engineering (QCE), 2026. arXiv:2604.18714.

Shifan Xu, Alvin Lu, and Yongshan Ding. “Fat-Tree QRAM: A High-Bandwidth Shared Quantum Random Access Memory for Parallel Queries.” Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Volume 2. 2025.

Samuel Stein, **Shifan Xu**, Andrew W. Cross, Theodore J. Yoder, Ali Javadi-Abhari, Chenxu Liu, Kun Liu, Zeyuan Zhou, Charles Guinn, Yufei Ding, Yongshan Ding, and Ang Li. “HetEC: Architectures for Heterogeneous Quantum Error Correction Codes.” Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Volume 2. 2025.

Shifan Xu, Connor T. Hann, Ben Foxman, Steven M. Girvin, and Yongshan Ding. “Systems Architecture for Quantum Random Access Memory.” In Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO), pp. 526–538. 2023.

PREPRINTS

Shifan Xu, Yizhuo Tan, Yongshan Ding, and Jakub Szefer. “OQRAM: Oblivious Quantum Random Access Memory for Securing Delegated Quantum Queries.” arXiv preprint arXiv:2607.27171 (2026).

Dantong Li, **Shifan Xu**, and Yongshan Ding. “Asymptotically Optimal Depth Fermionic Permutation on 2D Grid Quantum Architecture without Ancillas.” arXiv preprint arXiv:2605.26041 (2026).

Kun Liu, **Shifan Xu**, Tomas Jochym-O’Connor, Zhiyang He, Shraddha Singh, and Yongshan Ding. “In-Situ Simultaneous Magic State Injection on Arbitrary CSS qLDPC Codes.” arXiv preprint arXiv:2604.05126 (2026).

Adrian Harkness, Shuwen Kan, Chenxu Liu, Meng Wang, John M. Martyn, **Shifan Xu**, Diana Chamaki, Ethan Decker, Ying Mao, Luis F. Zuluaga, Tamás Terlaky, Ang Li, and Samuel Stein. “FTCircuitBench: A Benchmark Suite for Fault-Tolerant Quantum Compilation and Architecture.” arXiv preprint arXiv:2601.03185 (2026).

D. K. Weiss, **Shifan Xu**, Shruti Puri, Yongshan Ding, and Steven M. Girvin. “Faulty towers: recovering a functioning quantum random access memory in the presence of defective routers.” arXiv preprint arXiv:2411.15612 (2024).

Awards, Fellowships, & Grants

2020 **ZhengGang Fellowship**, Nanjing University

2019 **People Scholarship**, Nanjing University

Outstanding Prize in Taiwan College Physics Experiment tournament (TCPT), National Tsing Hua University

2018 **Outstanding Prize in Top-notch Talent Plan**, Nanjing University

Presentations

INVITED TALKS

Apr. 2026. ***Distilling Magic States in the Bicycle Architecture***. Invited talk hosted by the Xanadu team.

Spring 2024. ***Architectural Improvements in Quantum Random Access Memory***. Invited talk: Prof. Liang Jiang Group Meeting, University of Chicago.

Fall 2023. ***Systems Architecture for Quantum Random Access Memory***. Invited talk: QAISG's Quantum Machine Learning Seminar, Centre for Quantum Technologies, National University of Singapore.

Mentoring

2024-
present **Alvin Lu**, Undergraduate student, Yale University

New Haven

Outreach & Professional Development

SERVICE AND OUTREACH

2024 **Quantum Computing Applications and Systems Workshop (Co-located with 43rd ICCAD)**,
Program Committee

New Jersey,
USA

JOURNAL REFEREEING

Physical Review Research

Physical Review A (**PRA**)

PRX Quantum

Physical Review Letters (**PRL**)

Physical Review X (**PRX**)

IEEE Transactions on Parallel and Distributed Systems (**TPDS**)

ACM Transactions on Design Automation of Electronic Systems (**TODAES**)

ACM Transactions on Quantum Computing (**TQC**)

CONFERENCE REVIEWING

Quantum Information Processing (**QIP**)